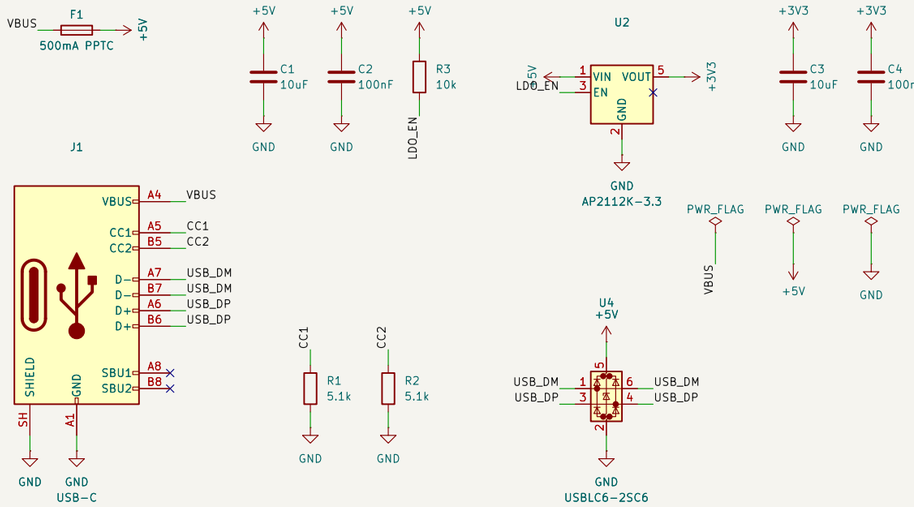
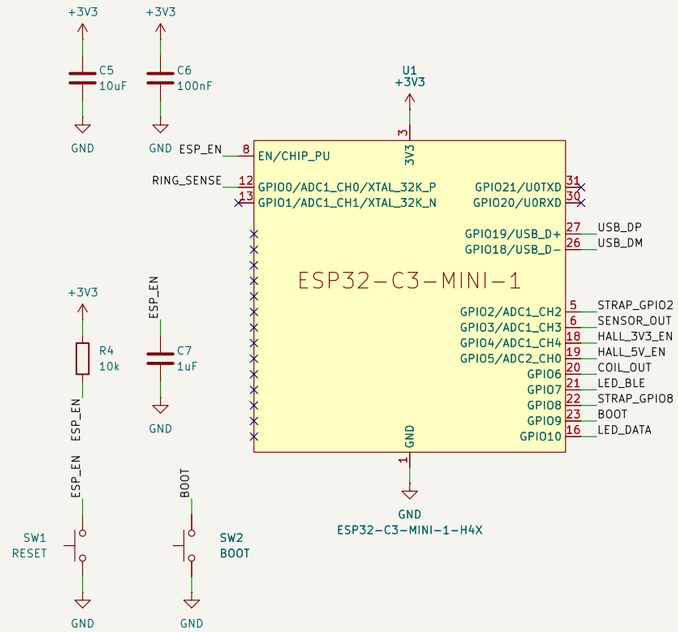
OpenRower sensor board – rev 2 (2026-10-17)

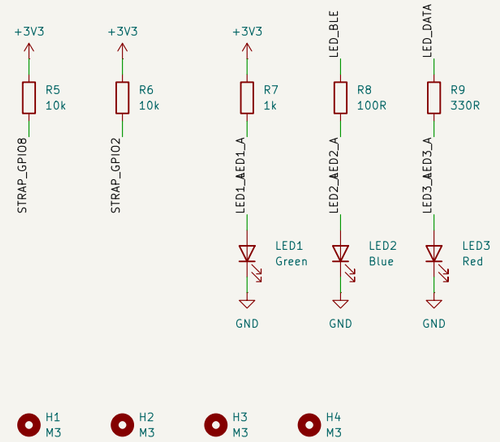
1. USB-C input, ESD, 3.3 V LDO



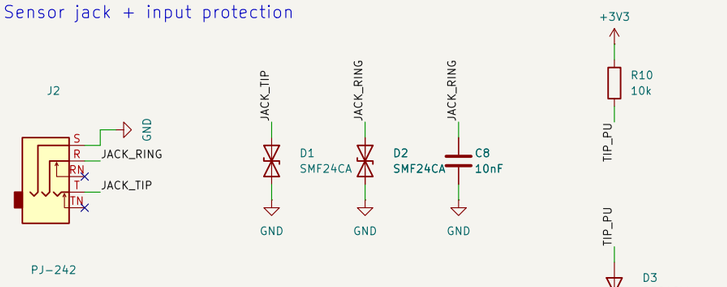
2. ESP32-C3-MINI-1-H4X (BLE + native USB CDC)



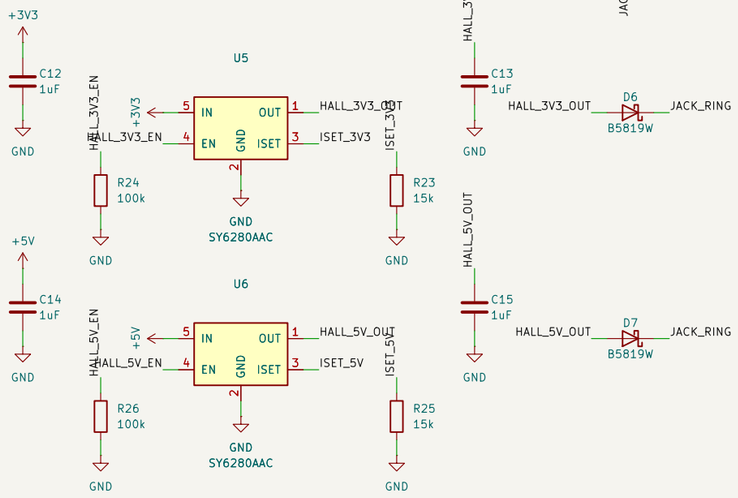
3. Strapping pull-ups, status LEDs, mounting holes



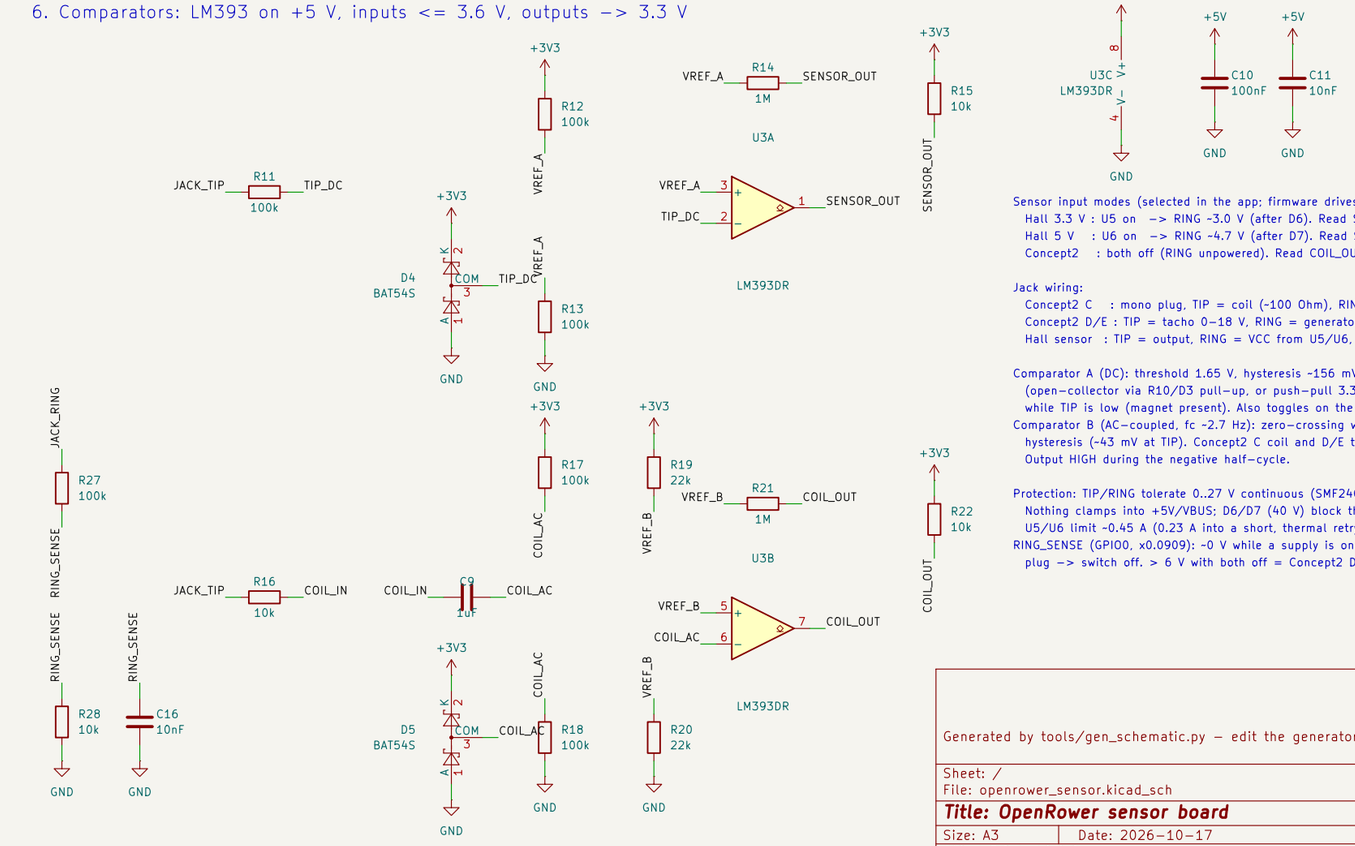
4. Sensor jack + input protection



5. Hall supply on RING (current-limited, reverse-blocking)



6. Comparators: LM393 on +5 V, inputs <= 3.6 V, outputs -> 3.3 V



Sensor input modes (selected in the app; firmware drives HALL_*_EN):
  Hall 3.3 V : U5 on -> RING ~3.0 V (after D6). Read SENSOR_OUT (GPIO3).
  Hall 5 V   : U6 on -> RING ~4.7 V (after D7). Read SENSOR_OUT (GPIO3).
  Concept2   : both off (RING unpowered). Read COIL_OUT (GPIO6).

Jack wiring:
  Concept2 C   : mono plug, TIP = coil (~100 Ohm), RING shorted to SLEEVE.
  Concept2 D/E : TIP = tacho 0-18 V, RING = generator DC ~18 V.
  Hall sensor : TIP = output, RING = VCC from U5/U6, SLEEVE = GND.

Comparator A (DC): threshold 1.65 V, hysteresis ~156 mV. Hall sensors
  (open-collector via R10/D3 pull-up, or push-pull 3.3/5 V). Output HIGH
  while TIP is low (magnet present). Also toggles on the D/E tacho.
Comparator B (AC-coupled, fc ~2.7 Hz): zero-crossing with ~36 mV
  hysteresis (~43 mV at TIP). Concept2 C coil and D/E tacho.
  Output HIGH during the negative half-cycle.

Protection: TIP/RING tolerate 0..27 V continuous (SMF24CA) in every mode.
  Nothing clamps into +5V/VBUS; D6/D7 (40 V) block the D/E generator.
  U5/U6 limit ~0.45 A (0.23 A into a short, thermal retry).
RING_SENSE (GPIO0, x0.0909): ~0 V while a supply is on = short or mono
  plug -> switch off. > 6 V with both off = Concept2 D/E generator.

Generated by tools/gen_schematic.py – edit the generator, not this file

Sheet: /		
File: openrower_sensor.kicad_sch		
Title: OpenRower sensor board		
Size: A3	Date: 2026-10-17	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/1